

N-QUEENS BACKTRACKING

N-Queens Backtracking Visualizer

N-QUEENS BACKTRACKING VISUALIZER

Interactive visualization of the Backtracking Algorithm

Number of Queens (N): 4

Start

Previous

Next Step

Auto Play

Reset

Chessboard

N-Queens Board (N = 4)

1



2



3



4



1

2

3

4

Algorithm Information

Solution Found!

Step: 48 / 78

Queen Positions: [3, 1, 4, 2]

Nodes Explored: 60

Solutions Found: 2

Algorithm Status

Solution 2 found!

All N queens have been placed without conflicts.

Complexity

Time: $O(N!)$

Space: $O(N)$

Technique: Backtracking

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Number of Queens (N): 4

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Auto Play

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Chessboard

N-Queens Board (N = 4)

1



2



3



4



1

2

3

4

Algorithm Information

Backtracking

Step: 32 / 78

Queen Positions: [2, 4, 1, 3]

Nodes Explored: 60

Solutions Found: 1

Algorithm Status

Backtrack: remove queen from Row 4, Column 3

The current choice did not lead to a solution. The algorithm returns to the previous decision and tries another column.

Complexity

Time: $O(N!)$

Space: $O(N)$

Technique: Backtracking

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Number of Queens (N): 4

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Chessboard

N-Queens Board (N = 4)

1

2

3

4

1

2

3

4

1

2

3

4

1

2

3

4

Algorithm Information

Backtracking

Step: 21 / 78

Queen Positions: [1]

Nodes Explored: 60

Solutions Found: 0

Algorithm Status

Backtrack: remove queen from Row 1, Column 1

The current choice did not lead to a solution. The algorithm returns to the previous decision and tries another column.

Complexity

Time: $O(N!)$

Space: $O(N)$

Technique: Backtracking

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Number of Queens (N): 4

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Chessboard

N-Queens Board (N = 4)

1

2

3

4

1

2

3

4

1

2

3

4

1

2

3

4

Algorithm Information

Backtracking

Step: 20 / 78

Queen Positions: [1, 4]

Nodes Explored: 60

Solutions Found: 0

Algorithm Status

Backtrack: remove queen from Row 2, Column 4

The current choice did not lead to a solution. The algorithm returns to the previous decision and tries another column.

Complexity

Time: $O(N!)$

Space: $O(N)$

Technique: Backtracking

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Chessboard

N-Queens Board (N = 4)

1

2

3

4

1

2

3

4

Algorithm Information

Invalid / Pruned

Step: 18 / 78

Queen Positions: [1, 4]

Nodes Explored: 60

Solutions Found: 0

Algorithm Status

Invalid placement at Row 3, Column 3 — branch pruned

The queen conflicts with another queen in the same column or diagonal. This branch is pruned.

Complexity

Time: O(N!)

Space: O(N)

Technique: Backtracking

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Number of Queens (N): 4

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Chessboard

N-Queens Board (N = 4)

1

2

3

4

1

2

3

4

Algorithm Information

Backtracking

Step: 17 / 78

Queen Positions: [1, 4, 2]

Nodes Explored: 60

Solutions Found: 0

Algorithm Status

Backtrack: remove queen from Row 3, Column 2

The current choice did not lead to a solution. The algorithm returns to the previous decision and tries another column.

Complexity

Time: O(N!)

Space: O(N)

Technique: Backtracking

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Chessboard

N-Queens Board (N = 4)

1

2

3

4

1

2

3

4

1

2

3

4

1

2

3

4

1

2

3

4

1

2

3

4

1

2

3

4

1

2

3

4

Algorithm Information

Invalid / Pruned

Step: 16 / 78

Queen Positions: [1, 4, 2]

Nodes Explored: 60

Solutions Found: 0

Algorithm Status

Invalid placement at Row 4, Column 4 — branch pruned

The queen conflicts with another queen in the same column or diagonal. This branch is pruned.

Complexity

Time: $O(N!)$

Space: $O(N)$

Technique: Backtracking

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N-Queens Board (N = 4)

1

2

3

4

1

2

3

4

1

2

3

4

1

2

3

4

1

2

3

4

1

2

3

4

1

2

3

4

1

2

3

4

Algorithm Information

Valid Placement

Step: 12 / 78

Queen Positions: [1, 4, 2]

Nodes Explored: 60

Solutions Found: 0

Algorithm Status

Place queen at Row 3, Column 2

The position is safe, so the algorithm recursively moves to the next row.

Complexity

Time: $O(N!)$

Space: $O(N)$

Technique: Backtracking

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Number of Queens (N): 4

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Chessboard

N-Queens Board (N = 4)

1

2

3

4

1

2

3

4

Algorithm Information

Valid Placement

Step: 10 / 78

Queen Positions: [1, 4]

Nodes Explored: 60

Solutions Found: 0

Algorithm Status

Place queen at Row 2, Column 4

The position is safe, so the algorithm recursively moves to the next row.

Complexity

Time: $O(N!)$

Space: $O(N)$

Technique: Backtracking

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Number of Queens (N): 4

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Chessboard

N-Queens Board (N = 4)

1

2

3

4

1

2

3

4

Algorithm Information

Backtracking

Step: 9 / 78

Queen Positions: [1, 3]

Nodes Explored: 60

Solutions Found: 0

Algorithm Status

Backtrack: remove queen from Row 2, Column 3

The current choice did not lead to a solution. The algorithm returns to the previous decision and tries another column.

Complexity

Time: $O(N!)$

Space: $O(N)$

Technique: Backtracking

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Chessboard

N-Queens Board (N = 4)

1

2

3

4

1

2

3

4

Algorithm Information

Invalid / Pruned

Step: 8 / 78

Queen Positions: [1, 3]

Nodes Explored: 60

Solutions Found: 0

Algorithm Status

Invalid placement at Row 3, Column 4 — branch pruned

The queen conflicts with another queen in the same column or diagonal. This branch is pruned.

Complexity

Time: $O(N!)$

Space: $O(N)$

Technique: Backtracking

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Number of Queens (N): 4

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Chessboard

N-Queens Board (N = 4)

1

2

3

4

1

2

3

4

Algorithm Information

Valid Placement

Step: 4 / 78

Queen Positions: [1, 3]

Nodes Explored: 60

Solutions Found: 0

Algorithm Status

Place queen at Row 2, Column 3

The position is safe, so the algorithm recursively moves to the next row.

Complexity

Time: $O(N!)$

Space: $O(N)$

Technique: Backtracking

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Chessboard

N-Queens Board (N = 4)

1

2

3

4

1

2

3

4

Algorithm Information

Invalid / Pruned

Step: 3 / 78

Queen Positions: [1]

Nodes Explored: 60

Solutions Found: 0

Algorithm Status

Invalid placement at Row 2, Column 2 — branch pruned

The queen conflicts with another queen in the same column or diagonal. This branch is pruned.

Complexity

Time: O(N!)

Space: O(N)

Technique: Backtracking

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Number of Queens (N): 4

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Chessboard

N-Queens Board (N = 4)

1

2

3

4

1

2

3

4

Algorithm Information

Completed

Step: 78 / 78

Queen Positions: [4]

Nodes Explored: 60

Solutions Found: 2

Algorithm Status

Backtrack: remove queen from Row 1, Column 4

The current choice did not lead to a solution. The algorithm returns to the previous decision and tries another column.

Complexity

Time: O(N!)

Space: O(N)

Technique: Backtracking